

Advanced (Habanero)

- Q1.** Solve the system $x + y = 4$ and $x - y = 2$ using substitution. What is x ?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q2.** Solve the system $2x + 3y = 7$ and $x - 2y = -3$ using elimination. What is y ?
- (A) -1
(B) 1
(C) 2
(D) 3
(E) 4
- Q3.** The system $x + y = 5$ and $2x + 2y = 10$ has
- (A) one solution
(B) no solution
(C) infinitely many solutions
(D) two solutions
(E) three solutions
- Q4.** The system $x - y = 1$ and $x + y = 3$ can be solved using
- (A) only substitution
(B) only elimination
(C) either substitution or elimination
(D) neither substitution nor elimination
(E) graphing only
- Q5.** Solve the system $x + 2y = 6$ and $3x - 2y = 4$ using elimination. What is x ?
- (A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q6.** The system $2x + 5y = 17$ and $x - 2y = -3$ has
- (A) one solution
(B) no solution
(C) infinitely many solutions
(D) two solutions
(E) three solutions
- Q7.** Solve the system $x - y = 2$ and $2x + 2y = 4$ using substitution. What is y ?
- (A) -2
(B) -1
(C) 0
(D) 1
(E) 2
- Q8.** The system $3x + 2y = 12$ and $x + 2y = 8$ can be solved using
- (A) only substitution
(B) only elimination
(C) either substitution or elimination
(D) neither substitution nor elimination
(E) graphing only

Open Ended Questions (FRQ)

- Q9.** Solve the system $x + 3y = 9$ and $2x - 2y = 4$ using elimination. Show all work and find x and y .
- Q10.** Solve the system $2x + y = 7$ and $x - 2y = -3$ using substitution. Show all work and find x and y .

Chef's Tips

- Emphasize the importance of checking the solution in both equations.
- Remind students to be precise with the signs of the coefficients.
- Encourage students to use the method that is most efficient for the given system.